

Python in Research Computation with Polynomials



Let's explore some of the common computational needs and how they can be met using Numpy.

Numpy is useful when complex mathematical computations are involved. This includes engineering problems for which Matlab, a very expensive product, is often used. See www.scipy.org/NumPy_for_Matlab_Users for more details. In this article, we will explore some common computational needs and look at how Numpy can address them.

Regression analysis / least square fit

A very common requirement is to have a model of a lot of collected data. You need to test whether the data fits the model. In this article, you can explore the Galilean model because, as mentioned in

www.vpri.org/pdf/rn2005001_learning.pdf, "This subject has been extensively studied by college students in the US: 70 per cent (including science majors) fail to understand this Galilean model of gravity near the surface of the Earth." It roughly means that a large proportion of the population believes that heavier objects fall faster. With the technology available these days, experimental testing of these beliefs is not difficult.

(Just for fun, check out the hammer and feather drop—Apollo 15 astronaut, David Scott, demonstrates on the Moon that Galileo was right, at en.wikipedia.org/wiki/Galileo%27s_Leaning_Tower_of_Pisa_experiment)